

Weighted Serrin Collar Reduction

This note fills in the “Input S” box from `selected-boundary-layer-theorem.md` and identifies the actual remaining sharpness obstruction.

The conclusion is:

1. the weighted D_H -to-Serrin upgrade is available in selected collars, at least in the standard oscillation form, by interpolation;
2. the specific Input S used in `selected-boundary-layer-theorem.md` is even simpler: it follows from quantitative isoperimetry through $D_I(t)$;
3. therefore the missing sharp piece is not Serrin stability. It is a boundary deficit propagation theorem that avoids the $1/\eta$ loss in the near-boundary slab average.

For clarity this note is volume-normalized: $|U| = |B_1| = \omega_n$. Constants depend only on n and on the selected-minimizer collar constants.

1 Selected-Collar Hypotheses

Assume $U = \{u > 0\}$ is a Plan 1 selected minimizer already in the graph regime. Thus there are $t_0 > 0$, $q_- > 0$, $q_+ < \infty$, and $M < \infty$ such that, for every regular $0 < t < t_0$,

$$\Sigma_t = \{u = t\}$$

is a $C^{1,\gamma}$ hypersurface with uniformly controlled charts, and

$$q_- \leq |\nabla u| \leq q_+ \quad \text{on } \Sigma_t, \quad \|\nabla u\|_{C^\gamma(\Sigma_t)} \leq M.$$

These are exactly the estimates supplied by Plan 1 graph entry, the Bernoulli condition, and Schauder estimates in the fixed collar.

2 Weighted Variance Gives Ordinary Variance

Let

$$f_t = |\nabla u|_{\Sigma_t}, \quad \bar{f}_t = \frac{1}{P(E_t)} \int_{\Sigma_t} f_t.$$

The level-set identity gives

$$\int_{\Sigma_t} \frac{(f_t - \bar{f}_t)^2}{f_t} = \frac{m(t)}{P(E_t)^2} D_H(t).$$

Since $q_- \leq f_t \leq q_+$ and $m(t), P(E_t) \simeq_n 1$ in the collar,

$$\|f_t - \bar{f}_t\|_{L^2(\Sigma_t)}^2 \leq C D_H(t).$$

Thus $D_H(t)$ is an ordinary L^2 Serrin defect on selected collars.

3 Interpolation to Oscillation

Let $d = n - 1$. On a uniformly $C^{1,\gamma}$ compact hypersurface Σ , if $g \in C^\gamma(\Sigma)$, then

$$\|g\|_{L^\infty(\Sigma)} \leq C \|g\|_{L^2(\Sigma)}^{\frac{2\gamma}{2\gamma+d}} [g]_{C^\gamma(\Sigma)}^{\frac{d}{2\gamma+d}}.$$

Applying this to $g = f_t - \bar{f}_t$, and using the uniform C^γ bound, gives

$$\|f_t - \bar{f}_t\|_{L^\infty(\Sigma_t)} \leq C D_H(t)^{\frac{\gamma}{2\gamma+n-1}}.$$

This is the precise bridge to Serrin-stability theorems stated in terms of oscillation of the normal derivative. It is not sharp as a power of D_H , but it is enough to show that good levels are genuine approximate Serrin domains.

4 Input S Is Already Implied by the Isoperimetric Deficit

The Input S used in `selected-boundary-layer-theorem.md` was

$$\mathcal{A}(E_t)^2 \leq C (D_H(t) + D_I(t) + |U \setminus E_t|^2),$$

in normalized units. This estimate does not require Serrin stability. The quantitative isoperimetric inequality gives directly

$$\mathcal{A}(E_t)^2 \leq C_n D_I(t),$$

because $m(t) \simeq_n 1$ in the selected collar. Hence Input S holds with D_H and the boundary-layer term unused:

$$\boxed{\mathcal{A}(E_t)^2 \leq C_n D_I(t) \leq C_n (D_H(t) + D_I(t) + |U \setminus E_t|^2)}.$$

So the weighted Serrin upgrade is useful diagnostic information, but it is not the bottleneck for Fraenkel asymmetry once the level set is already regular and near the ball.

5 Boundary Traces of the Level Deficits

The selected collar gives genuine boundary traces. Write

$$q(y) = |\nabla u(y)|, \quad y \in \partial U.$$

The normal expansion

$$\Sigma_t = \left\{ y - \frac{t}{q(y)} \nu_U(y) + O(t^{1+\gamma}) : y \in \partial U \right\}$$

implies

$$P(E_t) = P(U) + O(t^\gamma), \quad m(t) = |U| - t \int_{\partial U} \frac{1}{q} + O(t^{1+\gamma}),$$

and

$$\int_{\Sigma_t} |\nabla u| \rightarrow \int_{\partial U} q, \quad \int_{\Sigma_t} \frac{1}{|\nabla u|} \rightarrow \int_{\partial U} \frac{1}{q}.$$

Consequently

$$D_I(t) = D_I(0) + O(t^\gamma), \quad D_H(t) = D_H(0) + O(t^\gamma),$$

where

$$D_I(0) = P(U)^2 - c_n |U|^{2-2/n},$$

and

$$D_H(0) = \left(\int_{\partial U} \frac{1}{q} \right) \left(\int_{\partial U} q \right) - P(U)^2.$$

Thus the level-set deficits do pass to the free boundary. What is still missing is a sharp estimate of these boundary traces in terms of the torsion deficit.

6 The Tempting Boundary-Deficit Theorem Is Too Strong

The sharp boundary-layer theorem would follow from the following statement, but the statement is not true on the full nearly spherical class.

Boundary deficit propagation. For selected minimizers in the graph regime,

$$D_I(0) + D_H(0) \leq C(E(U) - E(B_1)).$$

If it held, choose

$$t \leq c(E(U) - E(B_1))^{1/\gamma}$$

inside the selected collar. Then

$$D_I(t) + D_H(t) \leq C(E(U) - E(B_1)),$$

and

$$|U \setminus E_t|^2 \leq Ct^2 = o(E(U) - E(B_1)).$$

Combining Input S with the transfer lemma gives

$$\mathcal{A}(U)^2 \leq C(E(U) - E(B_1)),$$

which is the sharp Saint-Venant/Faber-Krahn stability estimate for the selected minimizer.

The obstruction is spectral. For a spherical harmonic perturbation of degree $k \geq 2$, torsion energy is of size $k\|g_k\|_2^2$, while $D_I(0)$ and $D_H(0)$ are of size $k^2\|g_k\|_2^2$. Thus high frequencies violate the linear estimate.

7 Relation to Plan 1

The plausible replacement is the selected Bernoulli law. The Plan 1 notes on Bernoulli spectral closure linearize the Bernoulli condition and find a spectral multiplier $k - 1$ on spherical harmonics. This would suppress the non-neutral modes if the required tame $C^{2,\gamma} \times L^2$ Bernoulli remainder estimate were proved. That estimate is not currently established.

In boundary-deficit language:

- $D_H(0)$ measures failure of the Bernoulli datum $q = |\nabla u|$ to be constant.
- $D_I(0)$ measures the perimeter deficit of the selected free boundary.
- These defects are too strong to be bounded by torsion energy for arbitrary graphs. The Bernoulli spectral route is an attempted mechanism for excluding the high-frequency modes on the selected class, but it remains conditional on the missing tame expansion.

So the remaining work is either the BDV graph-entry plus nearly spherical second variation, or a fully justified Plan 1 Bernoulli spectral expansion. It is not a stronger Serrin theorem for rough interior levels.